

N SAI VIGHNESH

AI Engineer · Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate (9.3 CGPA) building AI-powered applications with hands-on experience in LLM integration, RAG pipelines (LangChain, ChromaDB, Ollama), and deep learning model training and deployment (PyTorch, DCGAN). Trained generative models and deployed them as live demos on Hugging Face Spaces with Gradio for real-time inference. Builds full-stack interfaces (React, Next.js, Spring Boot) to serve AI-driven features to end users. Proficient in containerization with Docker and CI/CD automation using Jenkins and Kubernetes for reproducible application deployment workflows. Oracle Cloud certified in Generative AI and DevOps. National-level hackathon participant (Meta PyTorch OpenEnv, Guidewire DEVTrails).

PROJECTS

TermBrain — Privacy-First AI Diagnostic Tool

Jan 2026 – Present

Python, Ollama, LangChain, ChromaDB, Typer, Rich, Docker, Jenkins, Kubernetes | [GitHub](#)

- Led a 3-person team to build an AI-powered CLI tool that diagnoses Linux and Kubernetes cluster errors using local LLM inference via Ollama, eliminating the need for cloud-based AI services.
- Implemented a RAG pipeline with LangChain and ChromaDB to index cluster documentation and log patterns, enabling context-aware error diagnosis with retrieval-augmented generation.
- Containerized the application with Docker and set up a Jenkins CI/CD pipeline for automated testing and deployment to a Kubernetes cluster.

GAN-Based Synthetic Image Generator

Jan 2026

Python, PyTorch, DCGAN, Gradio, Hugging Face Spaces | [Live Demo](#) | [GitHub](#)

- Implemented and trained vanilla GAN and DCGAN architectures in PyTorch to generate synthetic images of diseased plant leaves, augmenting limited training datasets for agricultural ML models.
- Compared training stability and output fidelity across multiple generator/discriminator configurations, documenting loss curves and sample quality at each training checkpoint.
- Deployed the generator as an interactive Gradio application on Hugging Face Spaces with CPU-optimized preprocessing for real-time inference.

ClipFlow AI — Developer Productivity Dashboard

Feb 2025

React, Next.js, PostgreSQL, AI/LLM Integration | [Live Demo](#) | [GitHub](#)

- Integrated an AI assistant that analyzes user-submitted algorithms and provides step-by-step hints and optimization suggestions.
- Built and deployed a full-stack productivity platform on Vercel that aggregates coding activity into a unified analytics dashboard with personalized streak tracking.
- Implemented server-side rendering via Next.js with PostgreSQL for persistent user data and session management.

Travel Planner — AI-Assisted Itinerary Builder

Dec 2025

React, Vite, Spring Boot, MySQL, Gemini API | [GitHub](#)

- Integrated the Gemini API to power a conversational chatbot that generates personalized day-wise travel schedules based on user preferences, budget, and travel dates.
- Built the full-stack serving layer with a React frontend and Spring Boot REST API backend, implementing prompt engineering and response parsing for structured AI-generated itinerary output.

TECHNICAL SKILLS

AI / ML: PyTorch, Scikit-learn, Ollama, LangChain, Gemini API

Languages: Python, Java, JavaScript, C, C++, SQL

LLM & Data: ChromaDB, RAG Pipelines, Gradio, NumPy, Pandas

Front-End: React.js, Next.js, HTML5, CSS3, Vite

DevOps & MLOps: Docker, Kubernetes, Jenkins, Hugging Face, Git

Back-End: Spring Boot, FastAPI, REST APIs, Supabase

EDUCATION

B.Tech, Computer Science & Engineering — KL University, Guntur, AP

Aug 2023 – May 2027

CGPA: 9.3 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | [Verify](#)

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | [Verify](#)

Oracle Cloud Infrastructure 2025 Certified DevOps Professional | [Verify](#)

HackerRank Software Engineer Role Certification | [Verify](#)

HACKATHONS & COMPETITIONS

- Meta PyTorch OpenEnv Hackathon x Scaler (Mar 2026)** — Developed robust Reinforcement Learning agent environments using Meta's OpenEnv framework for complex continuous control scenarios.
- Guidewire DEVTrails Hackathon 2026** — Built "GigShield," a 3-tier AI parametric insurance platform for Indian gig workers with ML-powered fraud detection and automated claim triggers using Scikit-learn.
- Design Thinking & Innovation @ KL-CIE (Dec 2023)** — Selected for Mock Shark Tank (Project Expo-3rd Edition) for "E-Waste Management" project, demonstrating design thinking and product pitch skills.